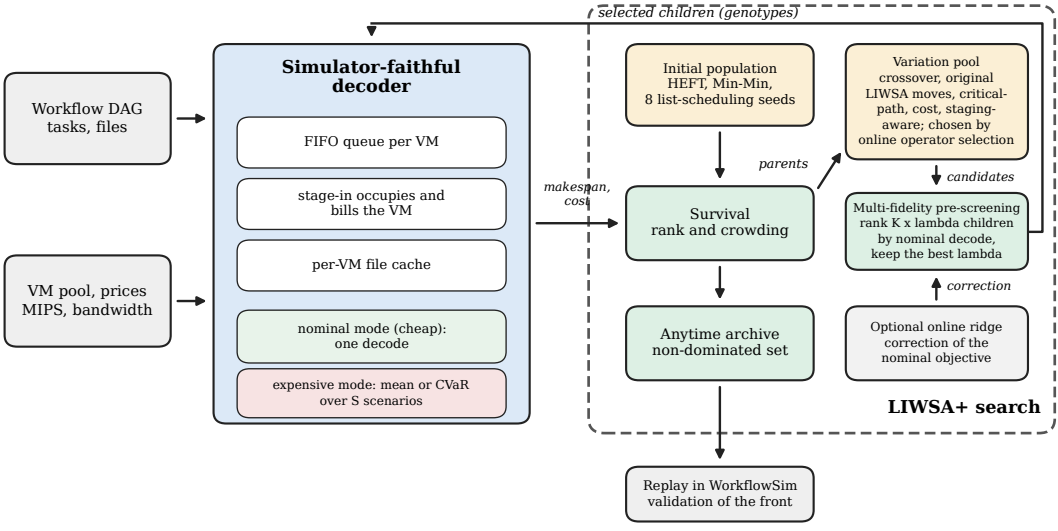
selected children (genotypes)
Workflow DAG
tasks, files
VM pool, prices
MIPS, bandwidth
Simulator-faithful
decoder
FIFO queue per VM
stage-in occupies and
bills the VM
per-VM file cache
nominal mode (cheap):
one decode
expensive mode: mean or CVaR
over S scenarios
makespan,
cost
Initial population
HEFT, Min-Min,
8 list-scheduling seeds
parents
Survival
rank and crowding
Anytime archive
non-dominated set
Variation pool
crossover, original
LIWSA moves, critical-
path, cost, staging-
aware; chosen by
online operator selection
candidates
Multi-fidelity pre-screening
rank K x lambda children
by nominal decode,
keep the best lambda
correction
Optional online ridge
correction of the
nominal objective
LIWSA+ search
Replay in WorkflowSim
validation of the front